

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

Student Name: J.Srikar

Roll No.: A24126552083

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TITLE

Implementation of JavaScript Arrays, Functions, and Object-Oriented Programming (Classes)

AIM

To understand and implement JavaScript arrays and ES6 classes by building a student marks calculator and a library management system.

OBJECTIVES

- Create and manipulate arrays using standard JavaScript methods.
- Define and invoke functions to perform iterative calculations (e.g., total and average).
- Implement an ES6 class with a constructor and member methods.
- Manage and manipulate arrays of objects within a class structure.

TASK DESCRIPTION

Design two JavaScript programs. The first program must manage an array of student marks using functions to display the array, calculate the total, calculate the average, and append new marks. The second program must establish a Library class to manage a collection of book objects, including methods to add new books, display the catalog, calculate the total cost of all books, and remove the most recently added book.

ALGORITHM/PROCEDURE

1. Array & Functions: Define an array of numbers representing student marks.

2. Create reusable functions to display the array, loop through the array to calculate the total sum, calculate the mathematical average, and push a new mark to the array.
3. Execute these functions and log the outputs to the console.
4. Library Class: Define a Library class with a constructor that initializes an empty books array.
5. Implement class methods: `addBook()` to push new book objects (title and cost), `displayBooks()` to iterate and print the catalog, `totalPrice()` to sum the costs, and `removeBook()` to pop the last book from the array.
6. Instantiate the Library class, add several books, and execute the methods to verify functionality.

PROGRAM/SOURCE CODE

Functions.js

```
// Array
let marks = [85, 90, 78, 88, 95];

// Function to display marks
function displayMarks(arr) {
  console.log("Student Marks: - function.js:6", arr);
}

// Function to calculate total
function totalMarks(arr) {
  let total = 0;

  for (let i = 0; i < arr.length; i++) {
    total += arr[i];
  }

  return total;
}

// Function to calculate average
function averageMarks(arr) {
  return totalMarks(arr) / arr.length;
}

// Function to add a new mark
function addMark(arr, mark) {
  arr.push(mark);
  return arr;
}

displayMarks(marks);

console.log("Total Marks: - function.js:33", totalMarks(marks));
```

```

console.log("Average Marks: - function.js:35", averageMarks(marks));

console.log("After Adding New Mark: - function.js:37");

addMark(marks, 92);

displayMarks(marks);

console.log("New Total: - function.js:43", totalMarks(marks));

```

Object.js

```

console.log("==== Library Management System ==== - object.js:1");

class Library {

  constructor() {
    this.books = [];
  }

  // Add Book
  addBook(name, price) {
    this.books.push({
      title: name,
      cost: price
    });

    console.log(name + "book added successfully. - object.js:16");
  }

  // Display Books
  displayBooks() {

    console.log("\nBooks in Library: - object.js:22");

    for (let i = 0; i < this.books.length; i++) {

      console.log(
        (i + 1) + ". " +
        this.books[i].title +
        " - ₹" +
        this.books[i].cost
      );
    }
  }
}

```

```

    }

    // Calculate Total Price
    totalPrice() {

        let total = 0;

        for (let i = 0; i < this.books.length; i++) {

            total += this.books[i].cost;

        }

        return total;

    }

    // Remove Last Book
    removeBook() {

        let removedBook = this.books.pop();

        console.log("\nRemoved Book: - object.js:57" + removedBook.title);

    }

}

const lib = new Library();

lib.addBook("Python", 500);
lib.addBook("Java", 600);

```

```

lib.addBook("HTML", 300);
lib.addBook("JavaScript", 400);

lib.displayBooks();

console.log("\nTotal Price: ₹ - object.js:72" + lib.totalPrice());

lib.removeBook();

console.log("\nAfter Removing One Book: - object.js:76");

lib.displayBooks();

console.log("\nNew Total Price: ₹ - object.js:80" + lib.totalPrice());

```

CODE EXPLANATION

JavaScript Element	Purpose & Explanation
function	Defines reusable blocks of code utilized here for mathematical operations like calculating total and average marks.
push()	A built-in array method used to append a new element (a mark or a book object) to the end of an array.

class	ES6 syntax used to define a blueprint (Library) for creating objects with shared properties and methods.
constructor()	A special class method that automatically executes upon object instantiation to initialize the empty books array.
pop()	A built-in array method that removes the last element from an array and returns it, used here to remove the latest book.
for loop	Used to iterate over the length of the arrays to accumulate totals and print catalog lists dynamically.

TEST CASES AND EXECUTION DATA

Test Case	Verification	Expected Result	Observed Result	Status
TC-01	Array sum calculation	Sum of [85, 90, 78, 88, 95] equals 436	Output returns 436 correctly	Pass
TC-02	Array element addition	addMark() appends 92 to the array	Array length increases; new total is 528	Pass
TC-03	Object instantiation	lib.addBook() pushes objects into the class array	4 books are added and displayed with their respective costs	Pass
TC-04	Class logic execution	lib.totalPrice() returns the sum of 4 books	Total calculates to 1800	Pass
TC-05	Object removal via pop()	removeBook() deletes the last added object	"JavaScript" is removed; new total updates to 1400	Pass

OUTPUT

```
Student Marks: - function.js:6 [ 85, 90, 78, 88, 95 ]
Total Marks: - function.js:33 436
Average Marks: - function.js:35 87.2
After Adding New Mark: - function.js:37
Student Marks: - function.js:6 [ 85, 90, 78, 88, 95, 92 ]
New Total: - function.js:43 528
```

```
===== Library Management System ===== - object.js:1
Pythonbook added successfully. - object.js:16
Javabook added successfully. - object.js:16
HTMLbook added successfully. - object.js:16
JavaScriptbook added successfully. - object.js:16

Books in Library: - object.js:22
1. Python - ₹500
2. Java - ₹600
3. HTML - ₹300
4. JavaScript - ₹400

Total Price: ₹ - object.js:721800

Removed Book: - object.js:57JavaScript

After Removing One Book: - object.js:76

Books in Library: - object.js:22
1. Python - ₹500
2. Java - ₹600
3. HTML - ₹300

New Total Price: ₹ - object.js:801400
```

RESULT

Thus, JavaScript concepts involving basic array manipulation, function definitions, and Object-Oriented ES6 classes were successfully implemented and tested. The console output verifies the accurate calculation of array logic and proper state management within the Library class.